

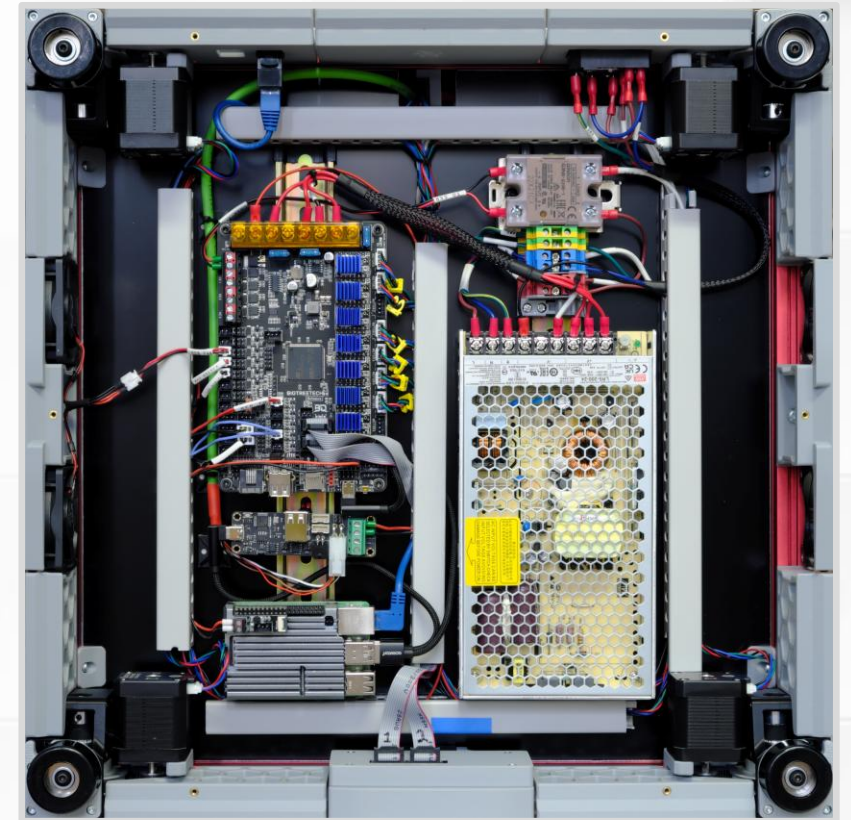
An Introduction to 3D Printing

Part 1: FFF Printers

*Process, Limitations, Dimensional Accuracy, and
Advantages*

About Me

- Woodworker.
- Learned 3D printing by building my own printer.
- Voron 2.4 with over 6k hours of print time.



([Voron 2.4 GitHub](#))

Websites

- 3D designs.
printables.com/FlyingGyroscope
- Get this guide.
github.com/FlyingGyroscope



Goals & Audience

- Learn *advantages* and *limitations* of 3D printing.
- Use printers effectively.
- Make great prints.
- Training guide for students.
- It assumes a teacher will setup, manage, and troubleshoot the printer.



Outline

Part 1: Printers

Process

Limitations

Dimensional Accuracy

Advantages

Part 2: Practical Application

Examples

Critical Thinking

Design

Part 3: Science

Vibrations and Waves

Linear Motion

Thermal Expansion

3D Printing is a Process

- 1) Start with a 3D design (digital file).
 - If you download, check the license for restrictions.
 - **STL**. Approximates shapes with tiny triangles.
 - **STEP**. Real geometry. Great for 3D modeling.
 - **3MF**. Compressed container, like a zip file.
Multiple objects, slicer settings, multi-color options, ...
- 2) Slice the design.
 - Printers do not accept 3D models.
 - Use software to “slice” objects into vertically-stacked layers.
 - Translate shapes into Gcode.

3D Printing is a Process

3) Print.

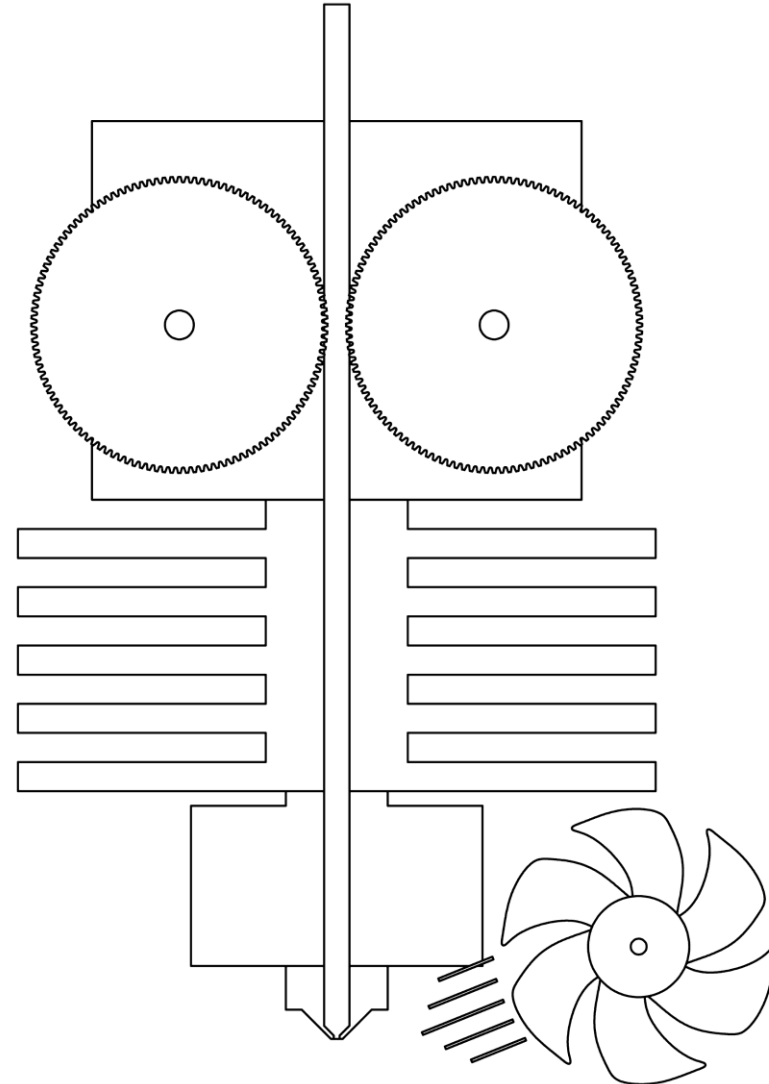
- Do you need to modify the design?
- Or tune printer settings?

4) Post-process.

- Remove supports.
- Assemble.
- Glue.
- Filler + sanding to remove layer lines.
- Paint.

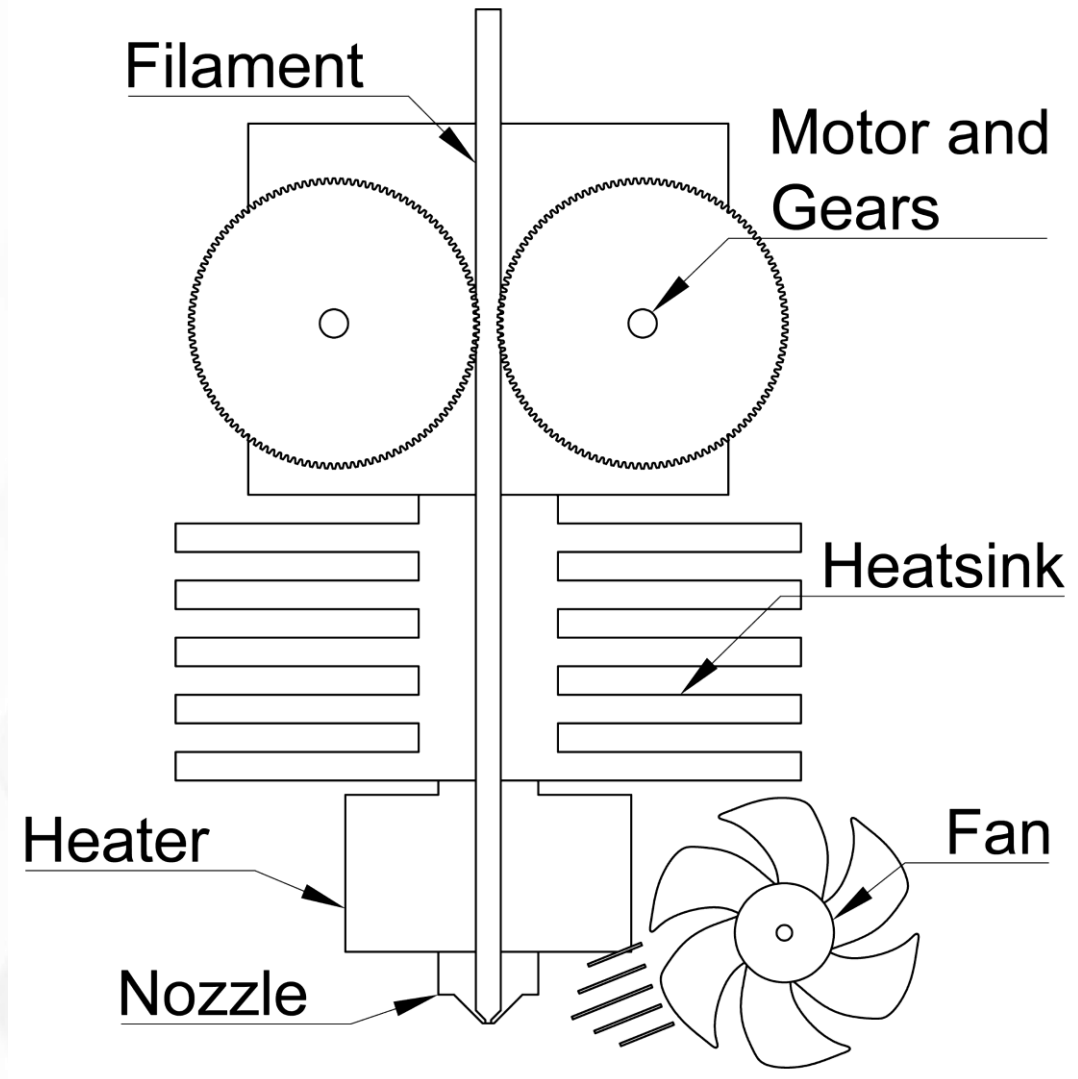
Pushing Plastic

- What kind of 3D printing?
- Filament and extrusion-based additive manufacturing.
- **FFF. Fused filament fabrication.**
- FDM™. Fundamentally the same but owned by Stratasys.
 - Fused deposition modeling.
- Toolhead. Moves around and deposits plastic.



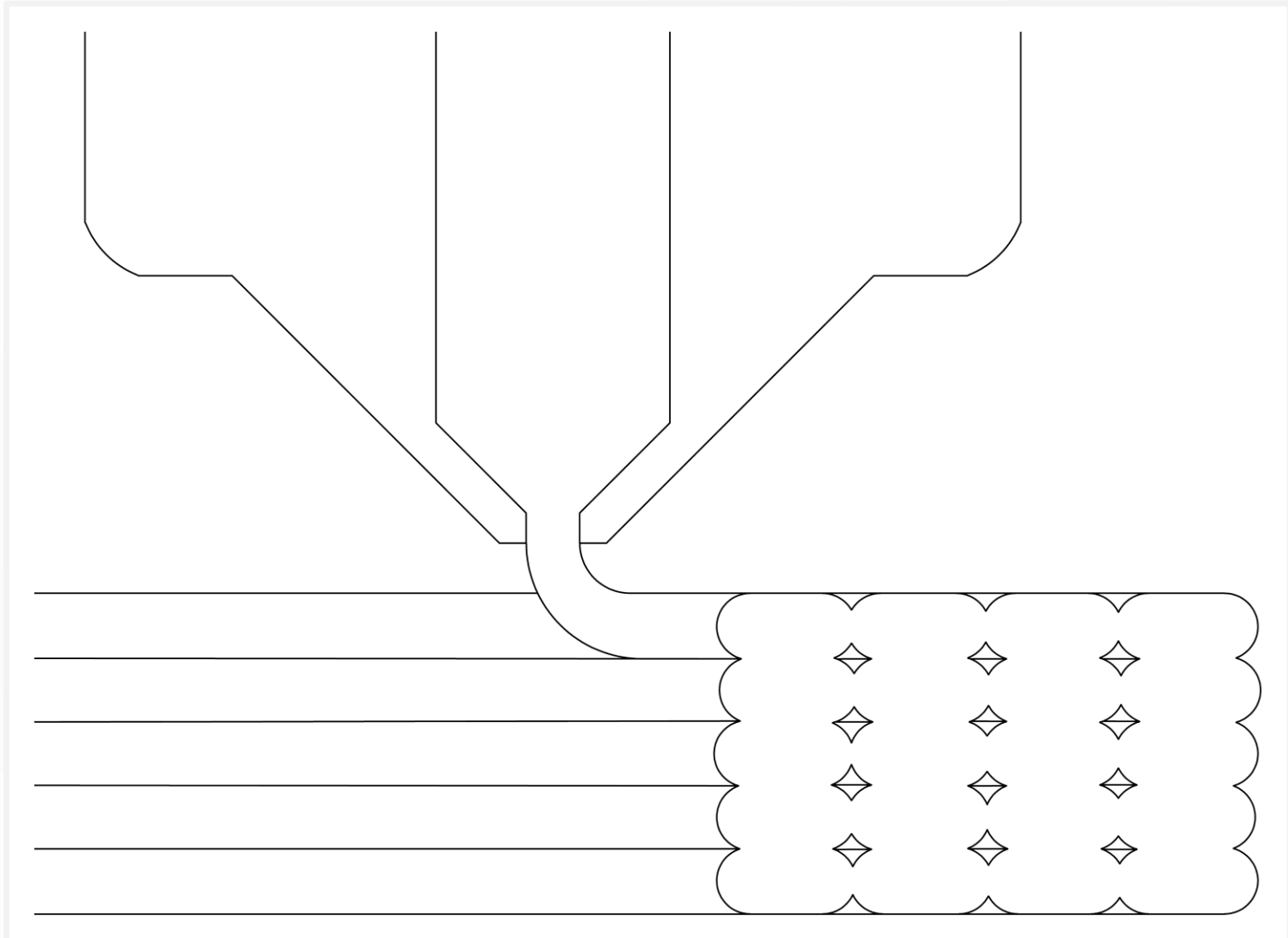
Pushing Plastic

- Filament. Long, continuous plastic rod.
- Motor and gears grab and push filament.
- Heatsink keeps heat in the hotend (heater + nozzle).
- Heater softens filament at the glass transition temperature.
- Nozzle opening shapes plastic.
- Fan cools hot plastic.



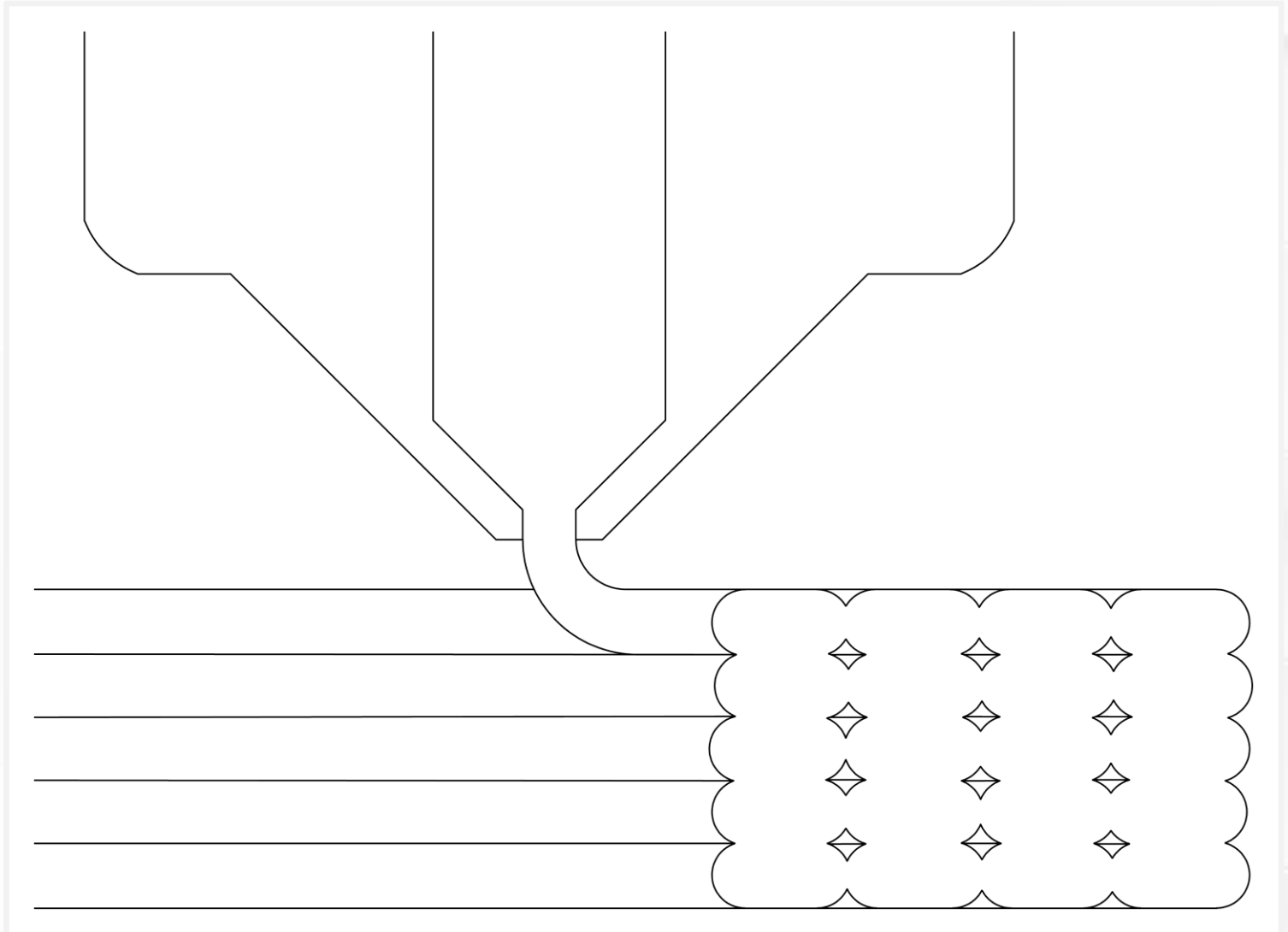
Extrusions

- Toolhead deposits lines called **extrusions**.
- Best to squish an extrusion into place.
- Lock in place as they cool.



Stacking Layers

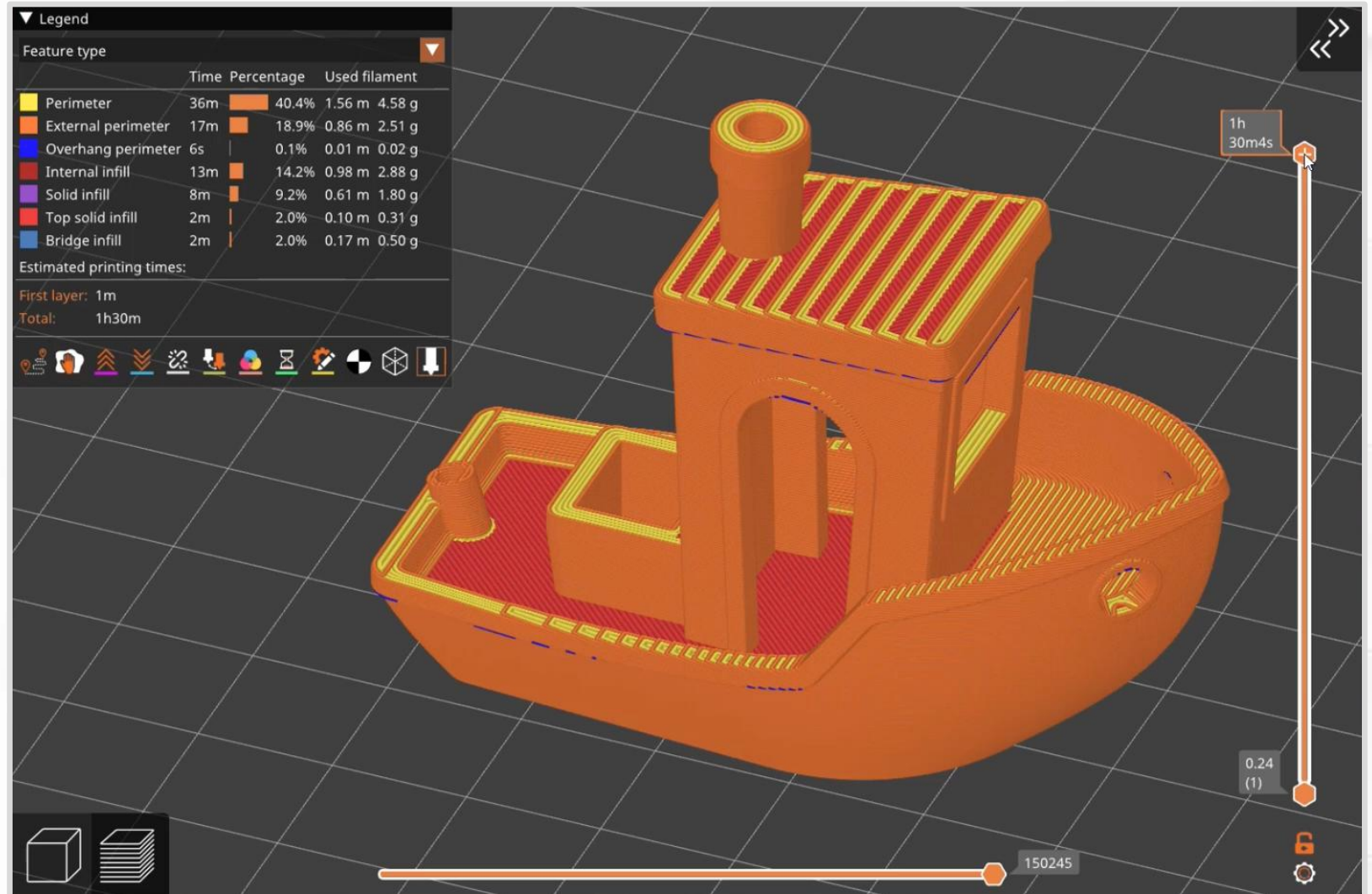
- Printers **simulate** 3D shapes by stacking layers.
- You could call it 2.5D printing.



Slicer Layers

Let's see how PrusaSlicer handles layers.

- This boat starts as a solid object.



(Animation,
[FlyingGyroscope GitHub](#))

Slicer Layers

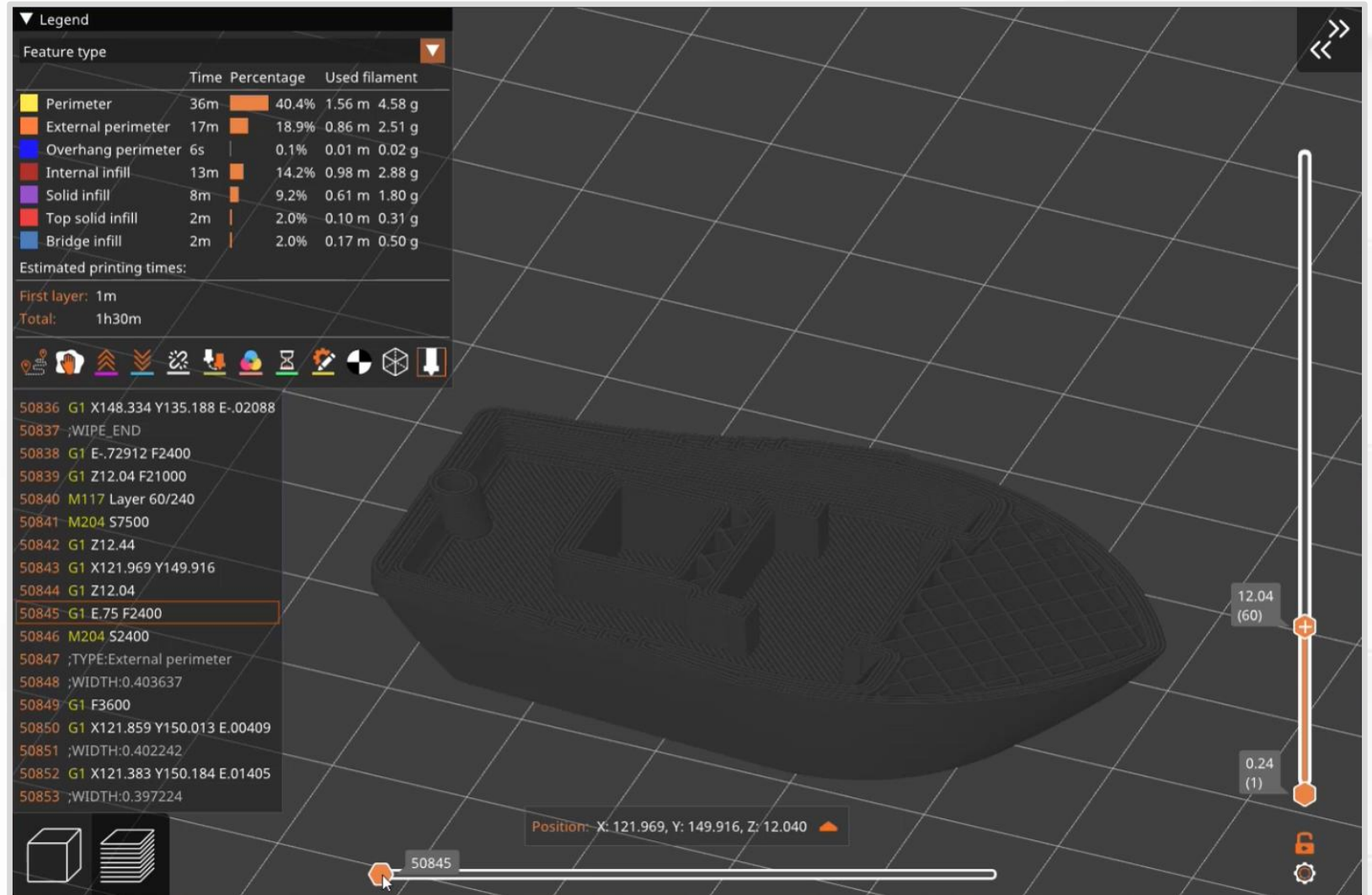
- Slicers sort extrusions by shape and geometry.
- Color-coded.
- Adjust settings for each category.

▼ Legend				
Feature type ▼				
	Time	Percentage	Used filament	
 Perimeter	36m	 40.4%	1.56 m	4.58 g
 External perimeter	17m	 18.9%	0.86 m	2.51 g
 Overhang perimeter	6s	 0.1%	0.01 m	0.02 g
 Internal infill	13m	 14.2%	0.98 m	2.88 g
 Solid infill	8m	 9.2%	0.61 m	1.80 g
 Top solid infill	2m	 2.0%	0.10 m	0.31 g
 Bridge infill	2m	 2.0%	0.17 m	0.50 g
Estimated printing times:				
First layer: 1m				
Total: 1h30m				

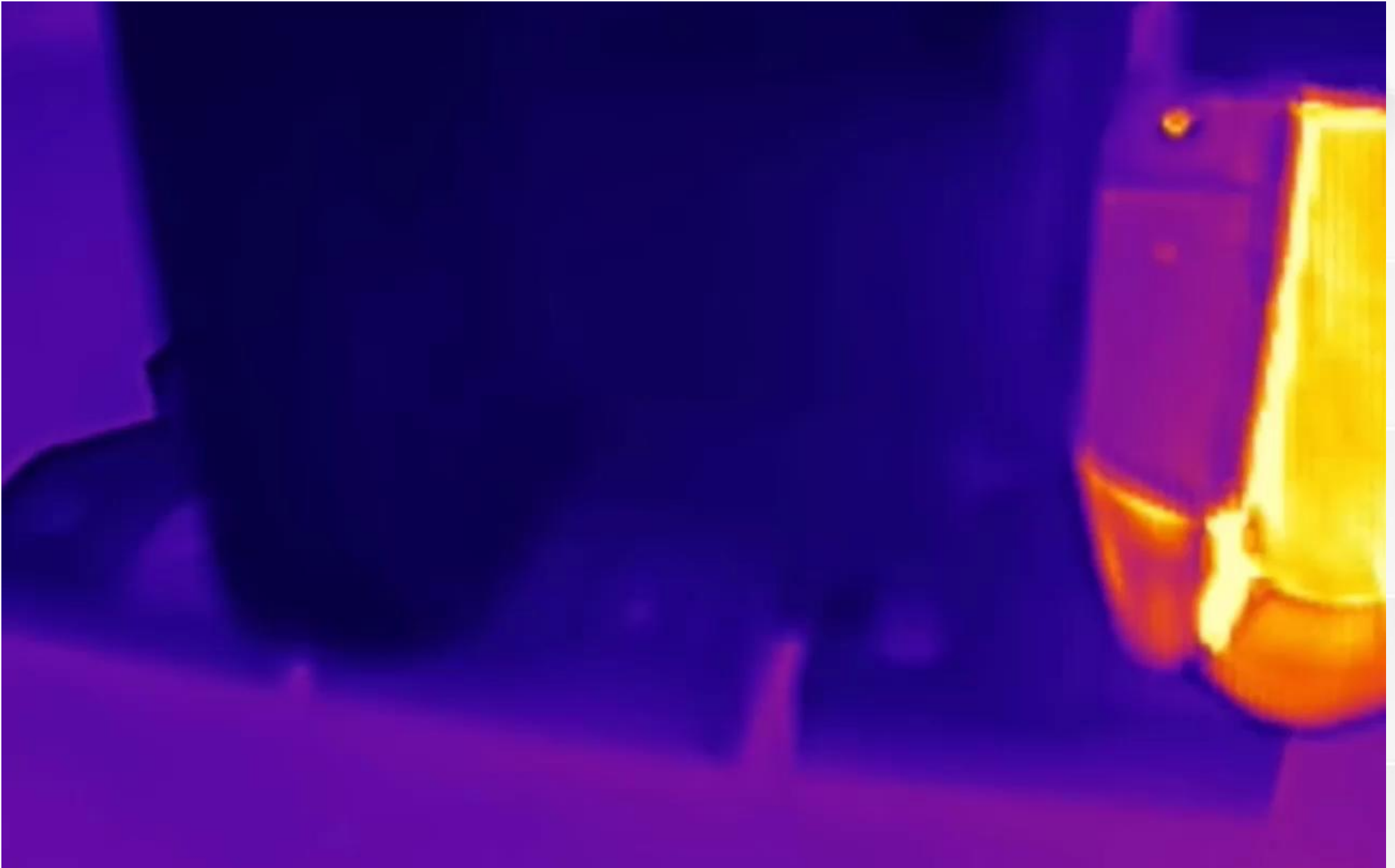
Slicer Layers

- Single layer preview.
- Gcode shows what the printer is actually doing (lower left).
- *G1* = move. Can include X, Y, or Z coordinates.
- *G1 X Y E* = move and extrude.

(Animation,
FlyingGyroscope GitHub)



(Animation,
Thermal camera
video by
NeedItMakelt)



Easy Filaments

- **PLA** (derived from corn starch).
 - General-purpose filament for indoor applications.
 - Excellent bridging.
 - Very stiff.
 - Deforms permanently under load (creep).
 - Low temperature resistance — will not survive inside a car on a hot summer day.
- **PETG** (plastic water bottles).
 - Pretty good general-purpose filament.
 - Hygroscopic (needs low-humidity storage).
 - Excellent choice for springs (elastic).
 - Moderate temperature resistance — can be used outdoors.

Easy Filaments

- What makes PLA and PETG easy?
 - Low warp (low shrink).
 - High dimensional accuracy.
 - Room temperature print chamber.

Proper Protocol

Let's review some everyday guidelines.

- Secure the loose end of a spool of filament at all times.
 - Tangles ruin prints and spools, and waste time.
- Do not reach into the printer while it is running.
 - Use the touchscreen or buttons to pause and resume prints.
- Watch the first layer before you walk away.
- Avoid burns.
 - Read nozzle and print bed temperatures on the display.
 - Never reach near a hot nozzle (210 °C = 410 °F).
- Do not leave fingerprints on print sheets.

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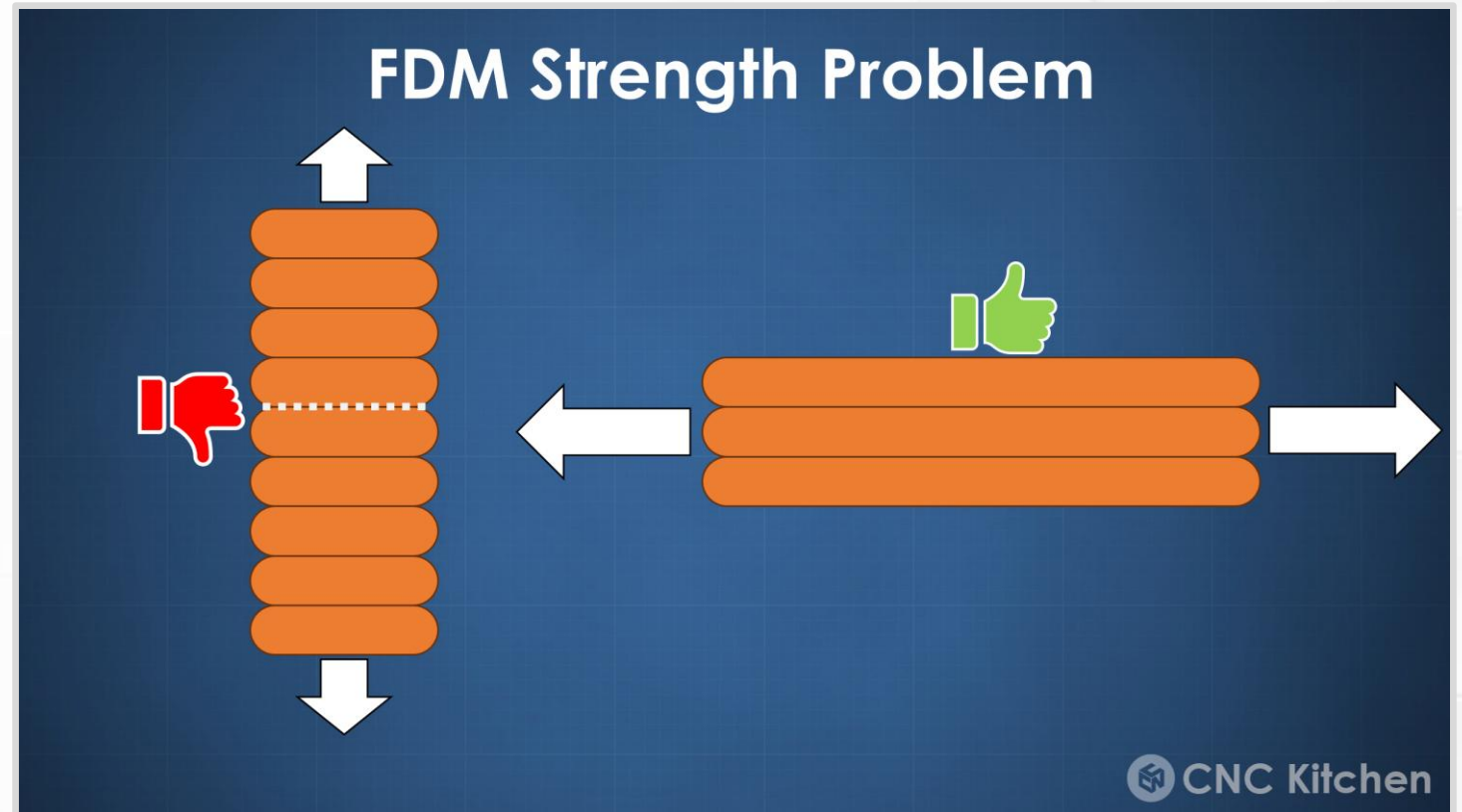
Thermal Expansion

Thermal Cycle

- Some plastics shrink as they cool.
 - Up to 1 or 2%.
 - Seems small, but large enough to distort dimensions.
 - Leads to cracking and delamination in extreme cases.
- Advice?
 - Tightly control ambient temperature and cooling rate.
 - Some filaments (ABS, Nylon) like it hot → Extra heating.
 - Some filaments (PLA) like it cold → Extra cooling.
 - PETG is somewhere in the middle.

Anisotropic Strength

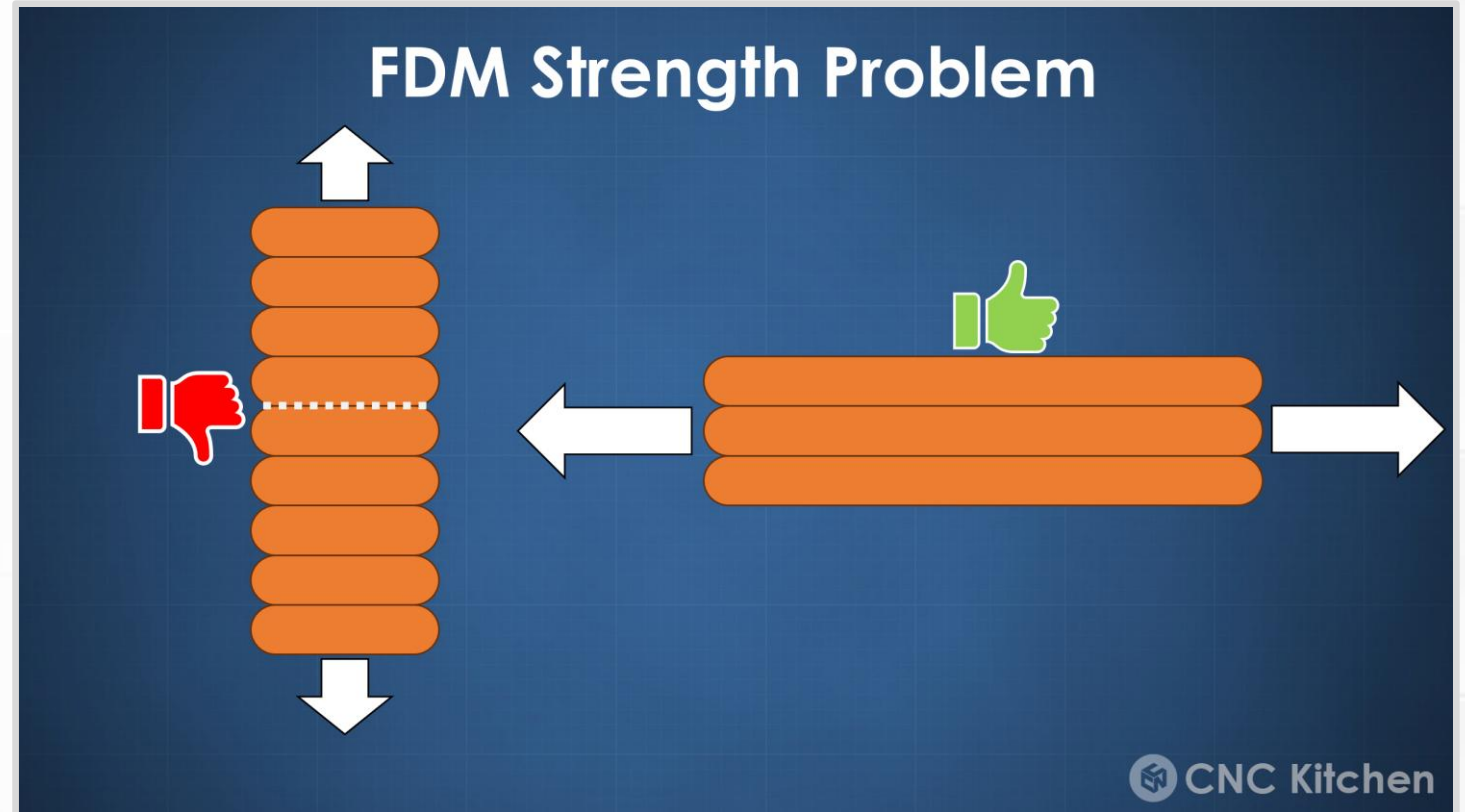
- Print orientation and layer lines affect strength.
- Layer-to-layer bonding is a weak point!
- Within an extrusion, hot filament bonds to hot filament.
- For a new layer, hot filament bonds to a warm layer.



(Picture from [CNC Kitchen](#))

Anisotropic Strength

- The “wrong” orientation can lose 50% of strength.
- Advice?
- Avoid tall and skinny shapes.
- Improve layer bonding.
 - Increase printing temps.
 - Decrease printing speed.



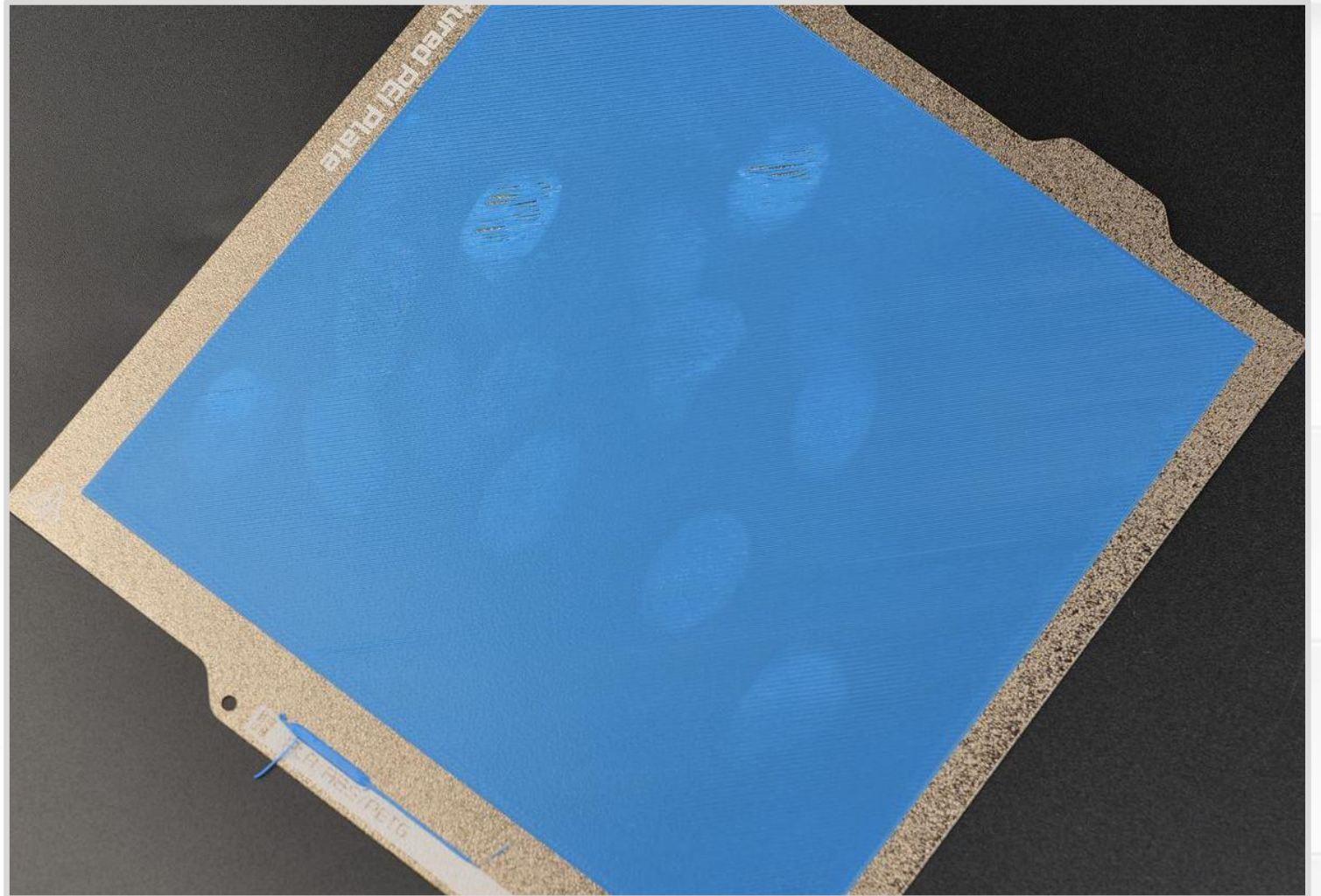
(Picture from [CNC Kitchen](#))

First Layer

- What is the most important layer in a 3D print?
 - The first layer!
 - First layers keep everything locked in place.
- Advice?
 - **Always watch the first layer.**
 - Tune first layer squish.
 - Flat surface area on the bottom of the 3D object.
 - Heated print bed.
 - Perfectly clean print bed.
 - Apply adhesion promoters (glue).

First Layer Mistakes

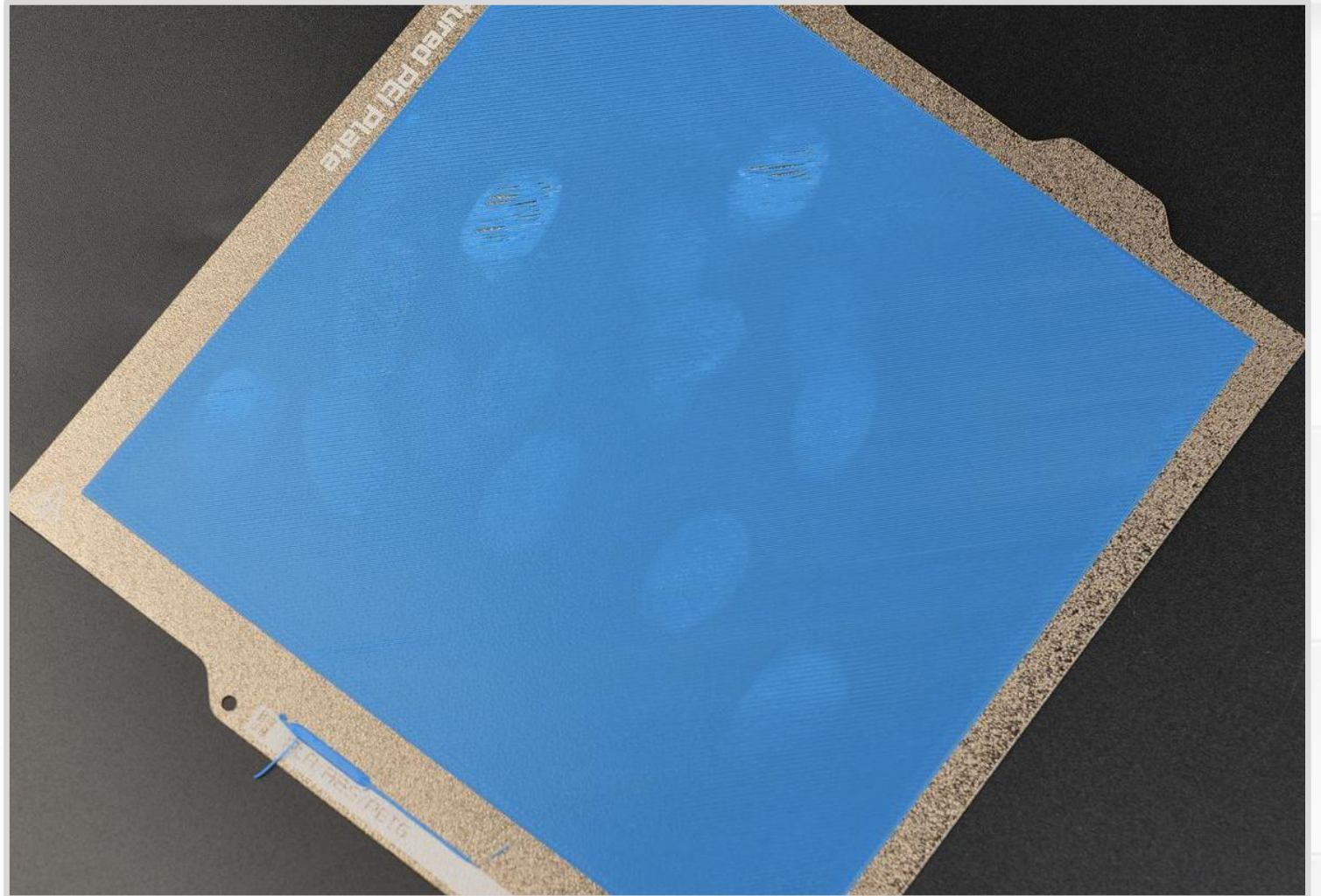
- What is wrong with this picture?
- **Fingerprints!** Plastic does not stick to oily fingerprints.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

- Advice?
- Deep cleaning: dish soap, soft brush, and water.
- Everyday use: window cleaner with ammonia.



(Picture from [Bambu Labs Wiki](#))

First Layer Mistakes

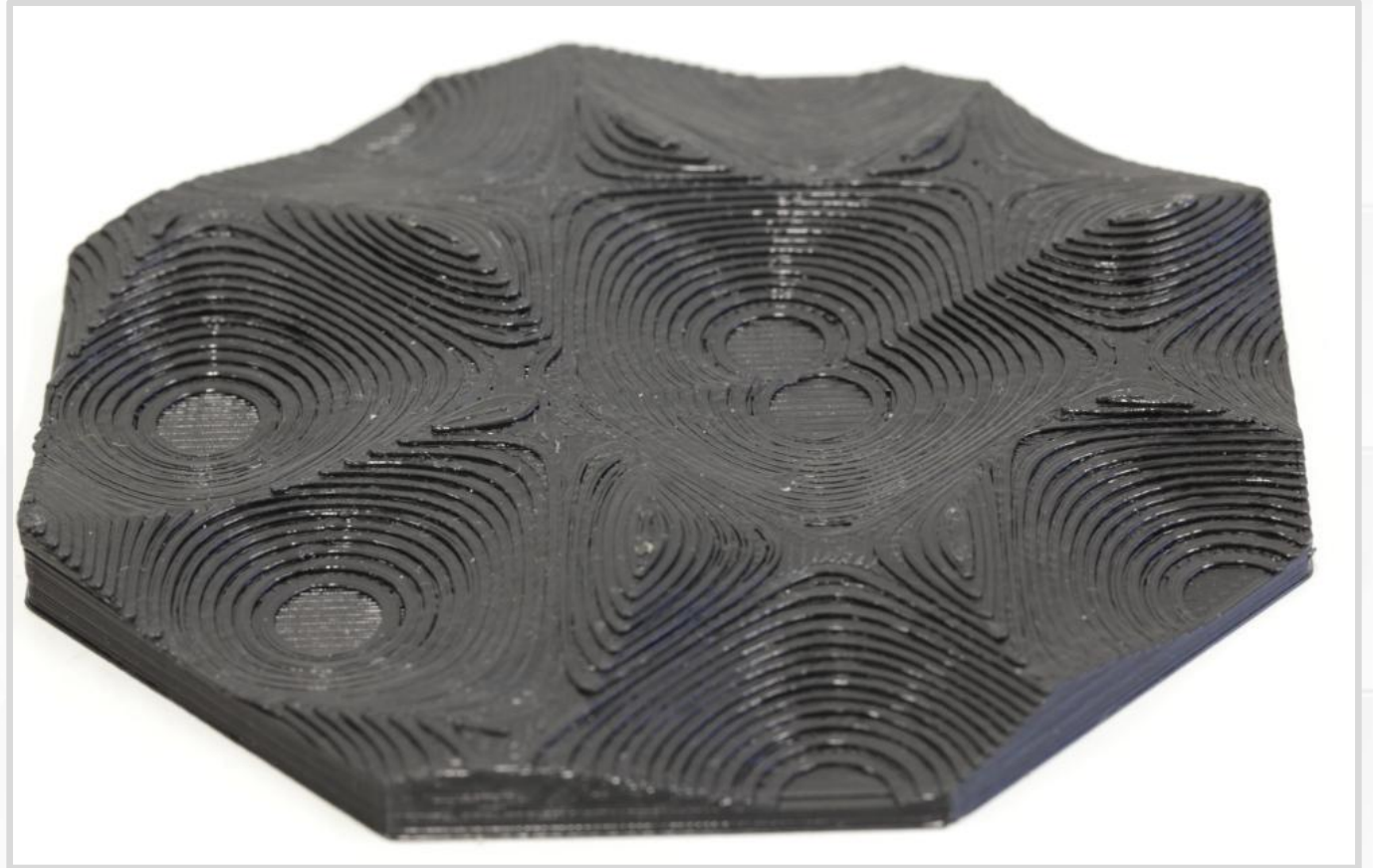
- What happens when a first layer does not stick?
- The print breaks free and makes “spaghetti”.
- You can see when things went wrong.
- Some of the rectangular print finished.



(Screenshot from Everything STEM)

Stair-Stepping

- Printing with layers → Vertical slopes turn into stair steps.
- Mostly cosmetic effect.
- More pronounced on shallow angles.
- Advice?
- Rotate objects to hide layer lines.
- Smaller layer height at the cost of increased print time.



(Picture from [Hamburg University Research Project](#))

First Layer Warping

- Why do corners and edges peel up?
- Regions shrink at different rates → Internal stresses → Stresses concentrate in corners.
- Shrinkage is a percent → Large parts have the most trouble.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

Consider this analogy.

- Imagine you want to remove a label or piece of tape.
- What do you do?
- Start peeling up the corner because that is the weak point.
- Same thing here.
- Physics found the weak point.



(Picture from [Creality Cloud Blog](#))

First Layer Warping

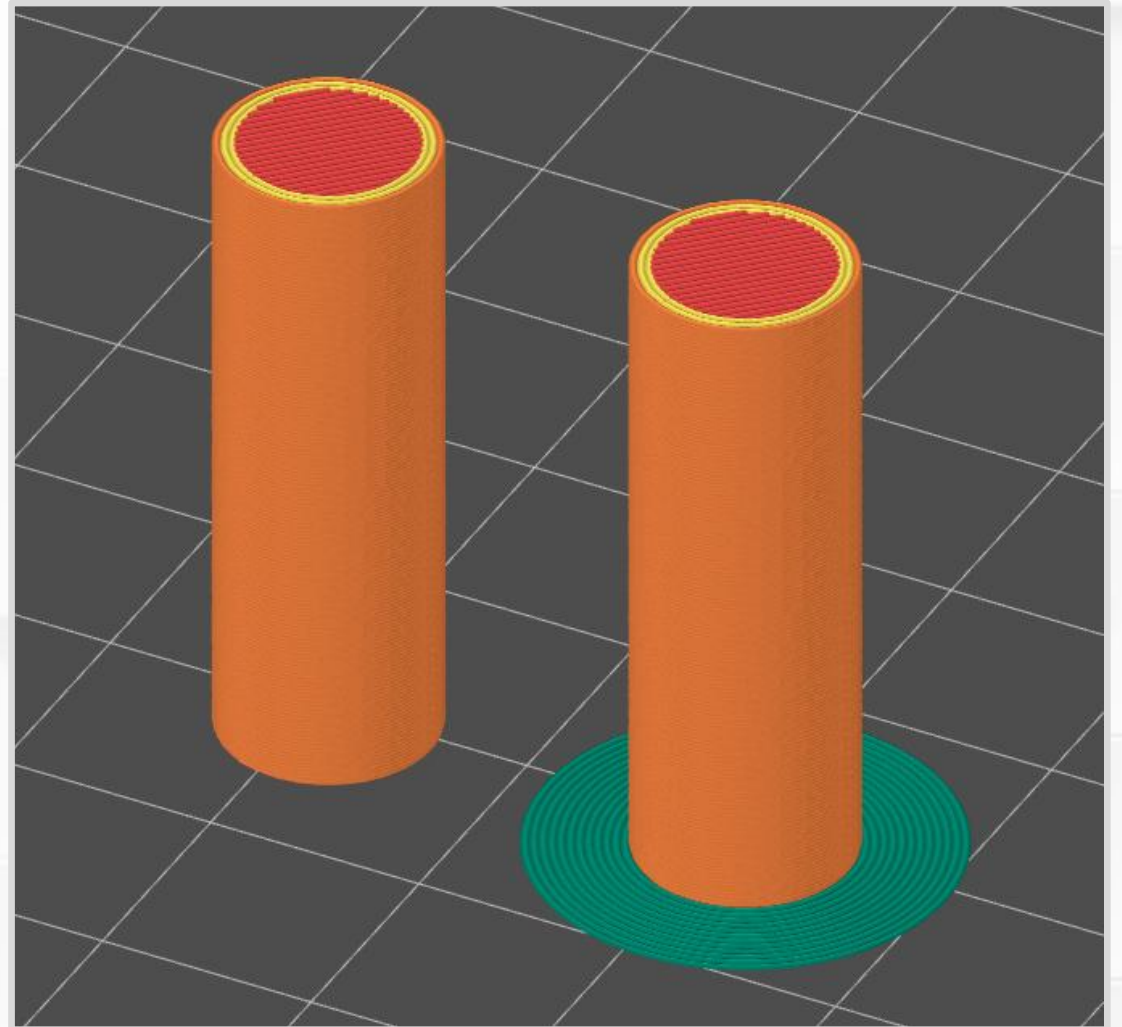
- Advice?
- Better control over temperatures.
 - Filament temperature.
 - Print chamber temperature.
 - Cooling fans.
- Evenly distributed cooling.
 - No fans or drafts that cool one side.
- Round corners to reduce stress concentration.



(Picture from [Crealitty Cloud Blog](#))

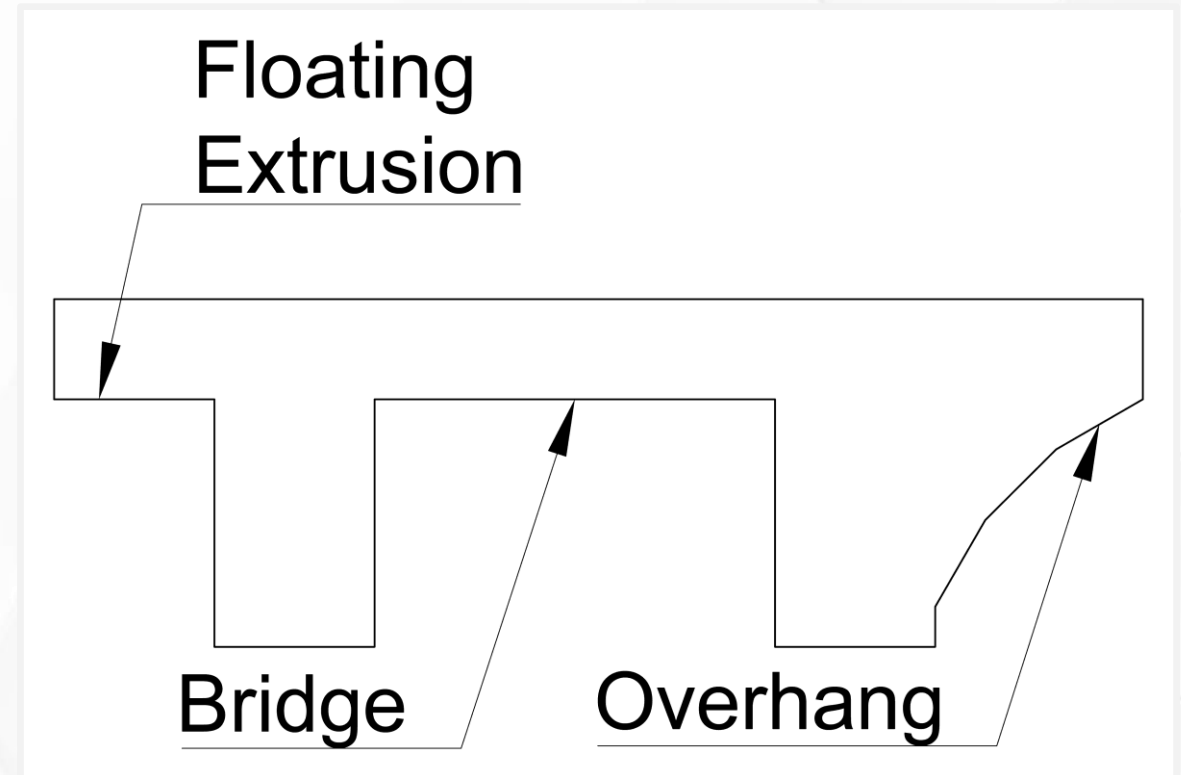
Brim

- Extra bottom surface area.
- Fights against first layer warping.
- Concentric extrusions joined to bottom perimeter. (Green lines on the right.)
- Added with a slicer.
- One layer tall.
- Easily peeled away after printing.



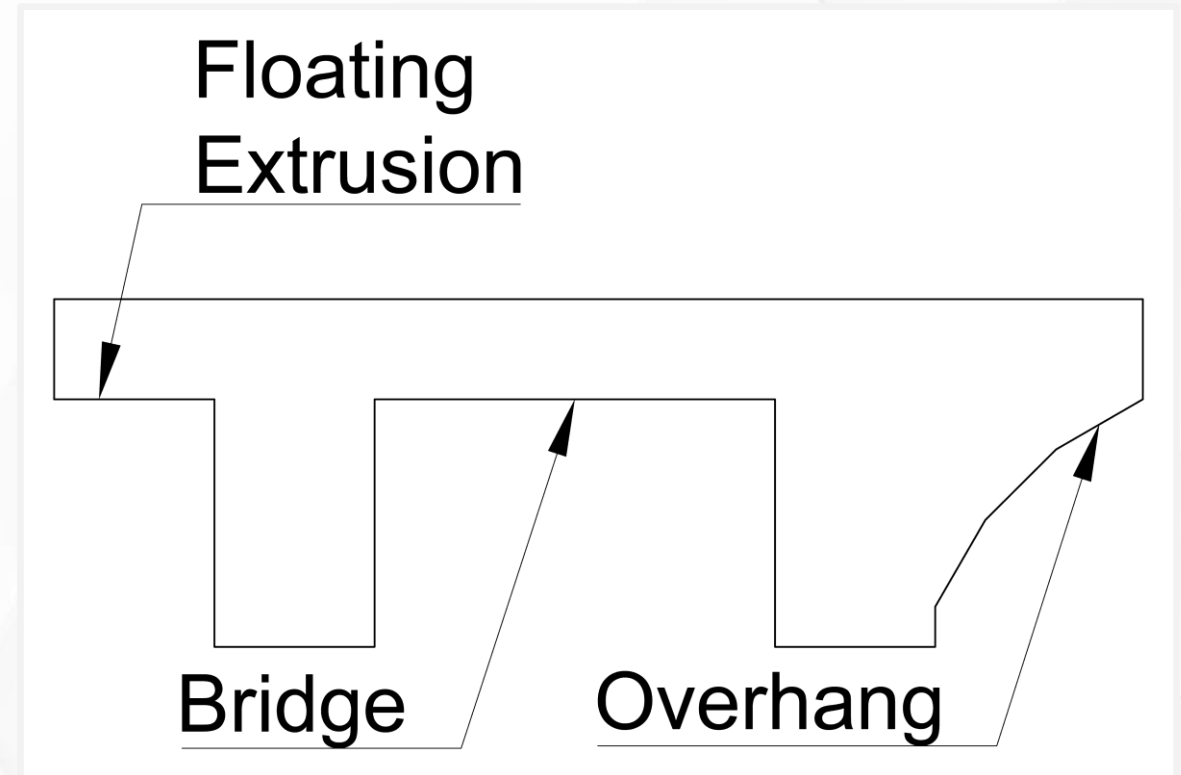
Levitating Extrusions

- *Critical thinking:* What happens when you try to levitate extrusions?
- Extrusions work best when you press them onto a solid surface.
- Predict what this print will look like.
- What parts will be the most deformed?



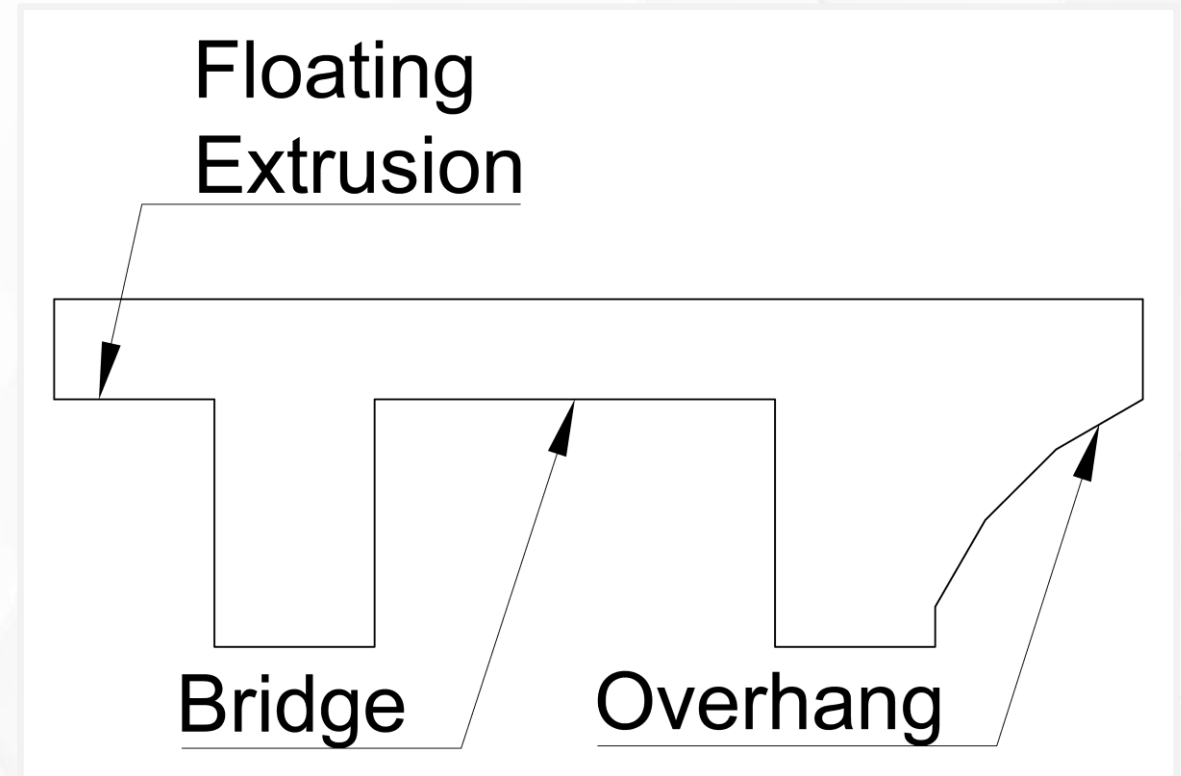
Overhang

- Extrusions that lean over an edge.
- Only the outermost extrusion is affected.
- Partially supported along one side.
- Advice?
- Keep overhangs at 45° to be safe.



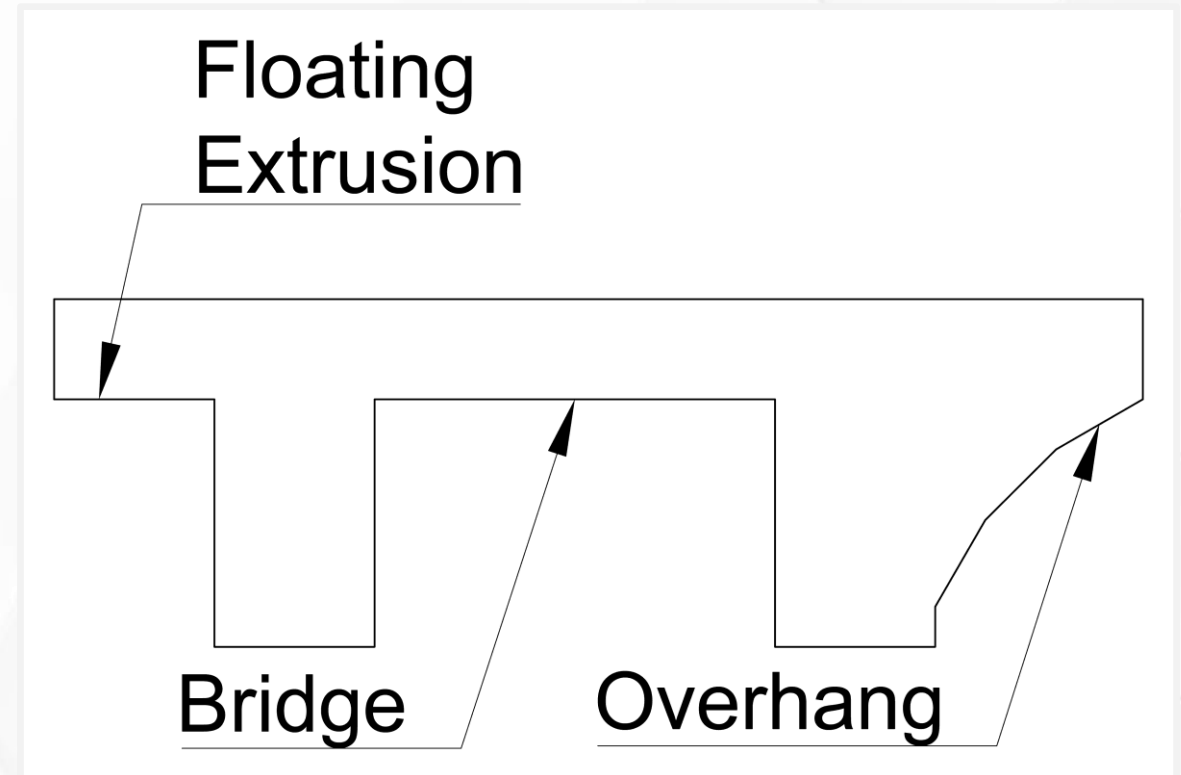
Bridge

- Both ends of an extrusion are supported and anchored.
- Middle hovers over a gap.
- The first few layers sag and droop down.
- Eventually recovers to flat layers.
- Advice?
- Keep bridges as short as possible.



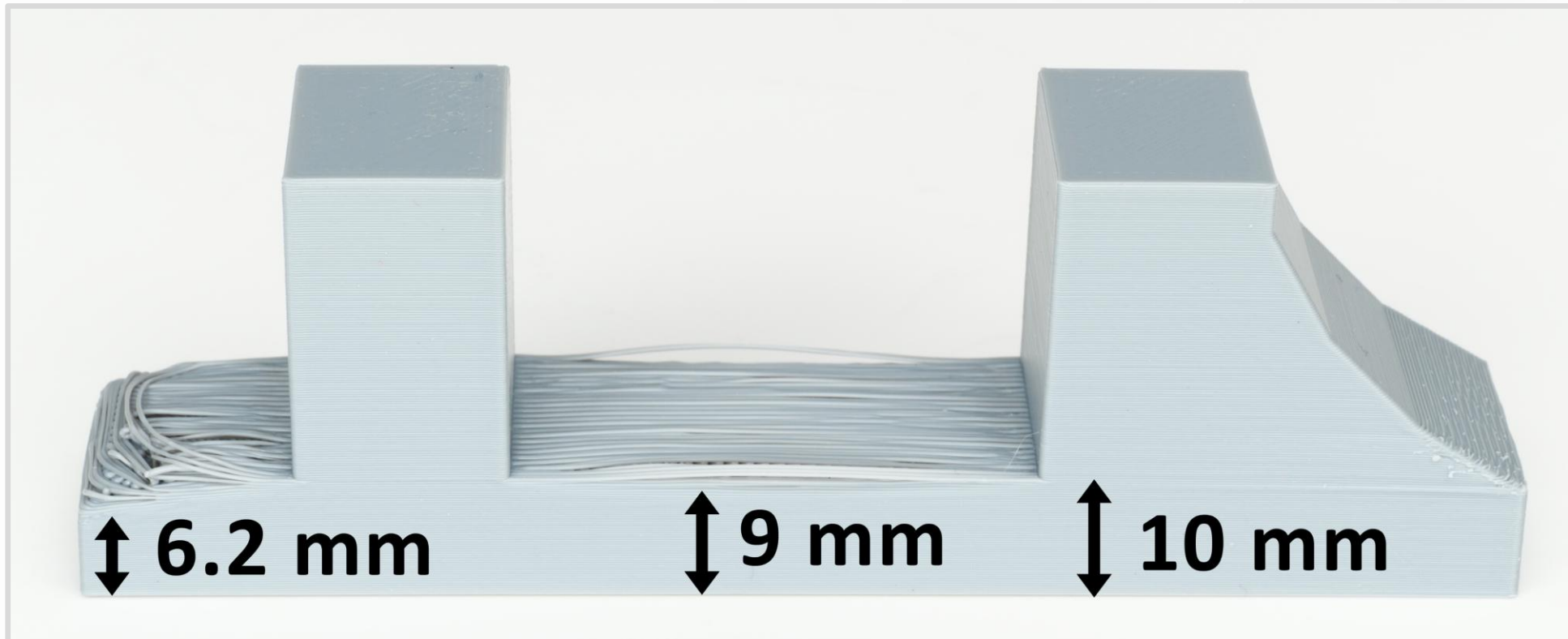
Floating Extrusion

- Extrusions that hang in the air.
- At least one end is not anchored to a solid surface.
- Advice?
- **Avoid. Very unreliable.**



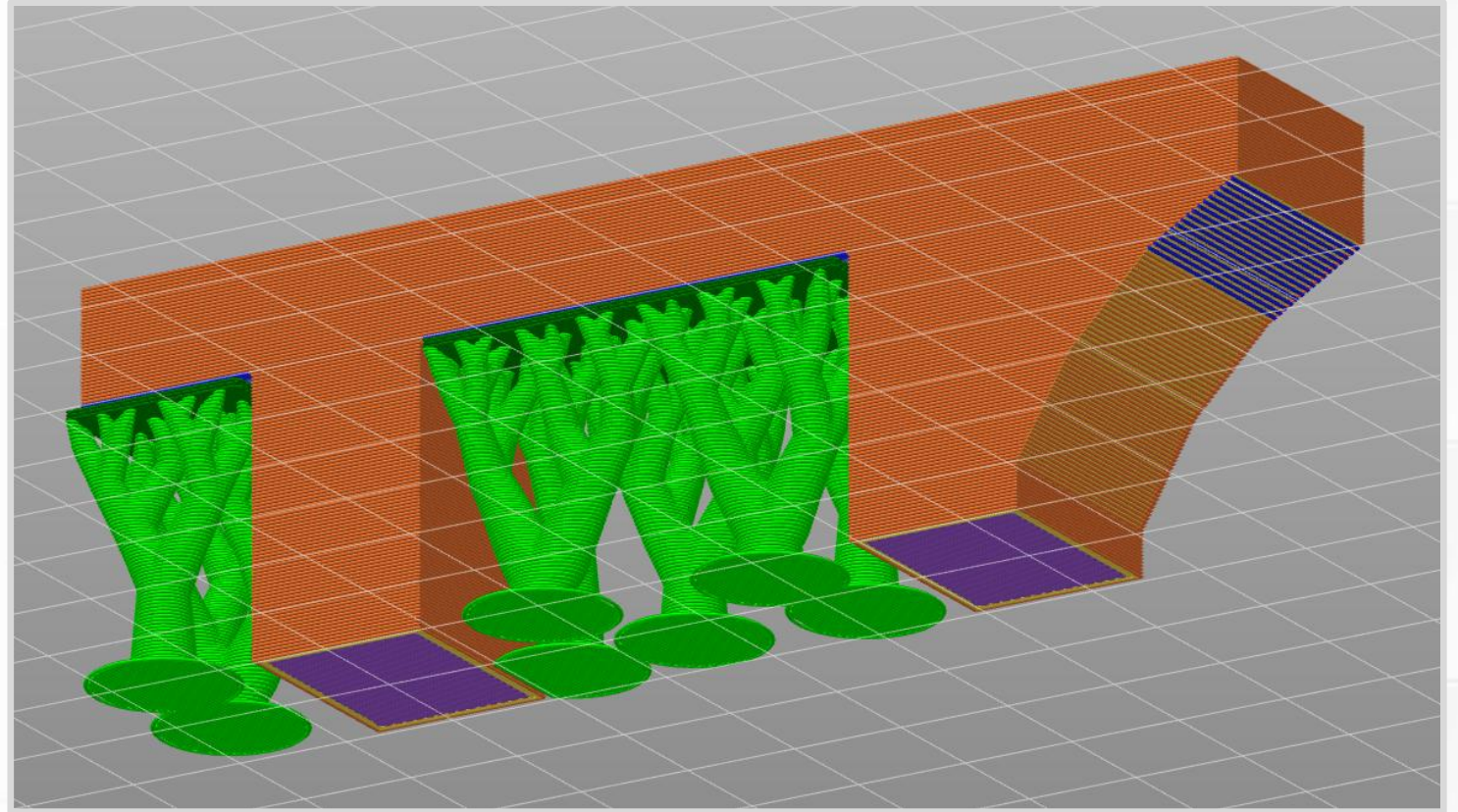
Stress Test Results

- All overhang angles look good.
- Bridge recovers to flat layers in 1 mm (**5 layers**).
- Floating extrusions recover in 3.8 mm (**19 layers**).



Supports

- Temporary scaffold that fixes:
 - Floating extrusions.
 - Long bridges.
 - Steep overhangs.
- Added by the slicer.
(Green trees-like branches.)
- Extra complexity, plastic, and print time.

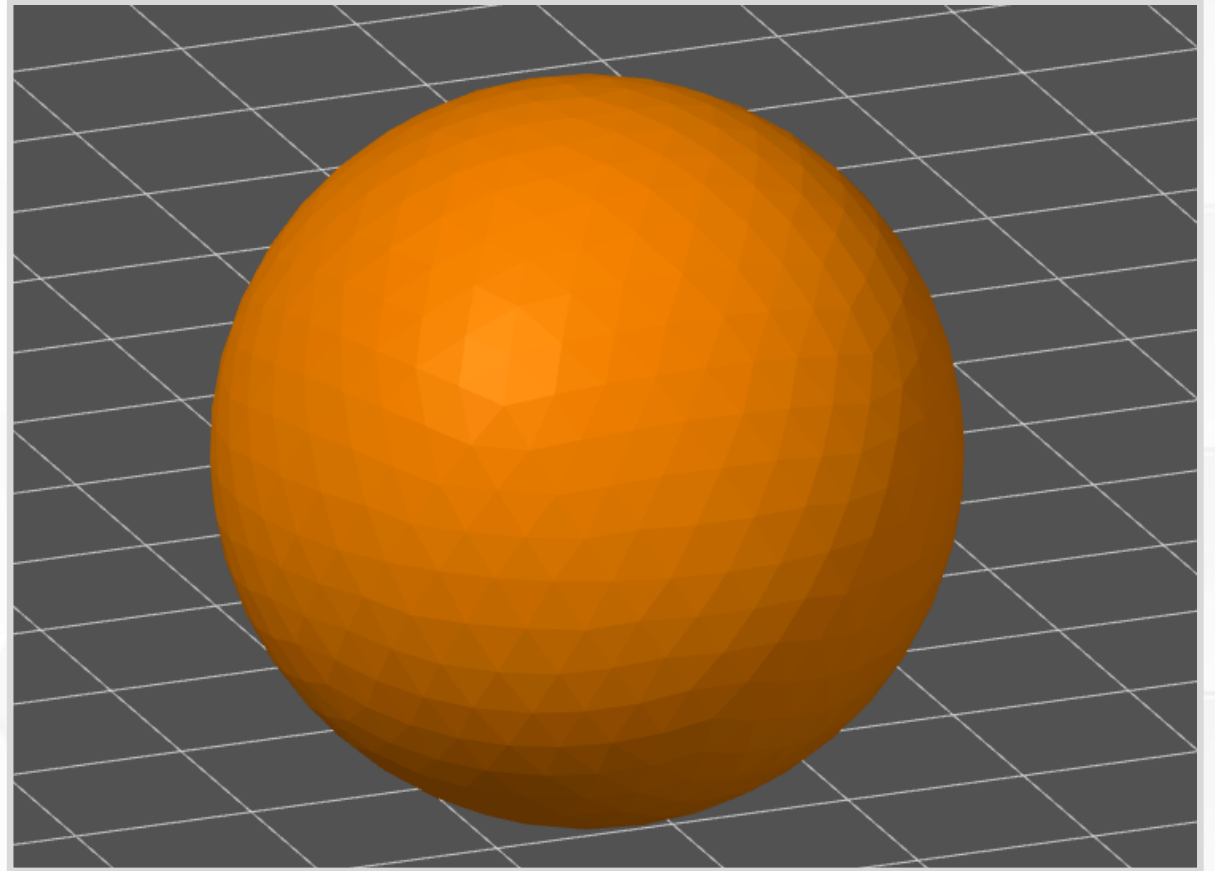


Supports

- In special situations, you can include supports as part of a 3D model.
- Tree/organic supports can reach into difficult spaces.
- Supports are a delicate balance. Why?
 - If they do not bond well enough ... they hold nothing in place.
 - If they bond too well ... they become part of the print.
- Supports can leave a rough surface finish.
- Sometimes you need to cut away pieces of support.

Limitations Review

- Printing a ball runs into some FFF limitations.
 - Very bad **first layer contact**. Huge spaghetti risk.
 - Steep overhangs on the bottom.
 - Second layer is steep enough to be a floating extrusion.
 - Stair-stepping creates uneven surface finish.



Limitations Review

How can we fix this print?

- Cannot fix stair-stepping by changing print orientation.
 - Use small layer heights to reduce visual artifacts.
 - Use adaptive layer heights to make the surface look more uniform.
- Brim is marginally helpful.
 - Object starts with too little surface area.
- Add supports for most of the bottom.
- Print two halves, then put them together.

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Dimensional Accuracy

- Do not design parts with exact dimensions.
- This is a trap for beginners.
- Why?
 - 3D prints are imperfect.
 - The fit is probably too tight to work as intended.
- Where do these errors come from?
- And how do we fix it?



Dimensional Accuracy

- Sources of error include (but are not limited to):
 - Shrinking and warping.
 - Inconsistent extrusions.
 - Misaligned parts.
 - Bent parts from shipping damage.
 - Incorrect belt tension.
 - Backlash, slop, or loose parts.
 - Moving parts that eventually wear down.
 - Unoptimized slicer settings.
 - Thermal expansion (discussed in Part 3).
 - Vibrations (discussed in Part 3).

Modeling Calibration

- *(Printed size) = (Size) × (Scale factor) + (Constant offset).*
- Simplified model.
 - Look at one dimension at a time.
 - Assume fully functional motion system.
- Scale factor. Error changes with size.
 - Part shrinkage.
- Constant offset. Error is always the same.
 - Extrusion inconsistencies. Over and under-extruding.
 - Surface roughness. Additives like carbon fiber, wet filament.

Scale Factor

- Useful in certain scenarios.
 - Large parts (over 6 inches, 150 mm).
 - Filaments that shrink (like ABS).
- Fixed by adjusting size-related settings in your slicer.
 - XY part scaling.
 - Shrinkage compensation.

Constant Offset

- Very important for any project with assembled pieces!
- **Add clearance** (air gap) between parts that touch.
- Best to add clearance while designing 3D objects.
- Also try adding negative XY offset in slicer.
 - Equally offset all vertical surfaces.
- I consider XY clearance in all of my designs.

Adding XY Clearance

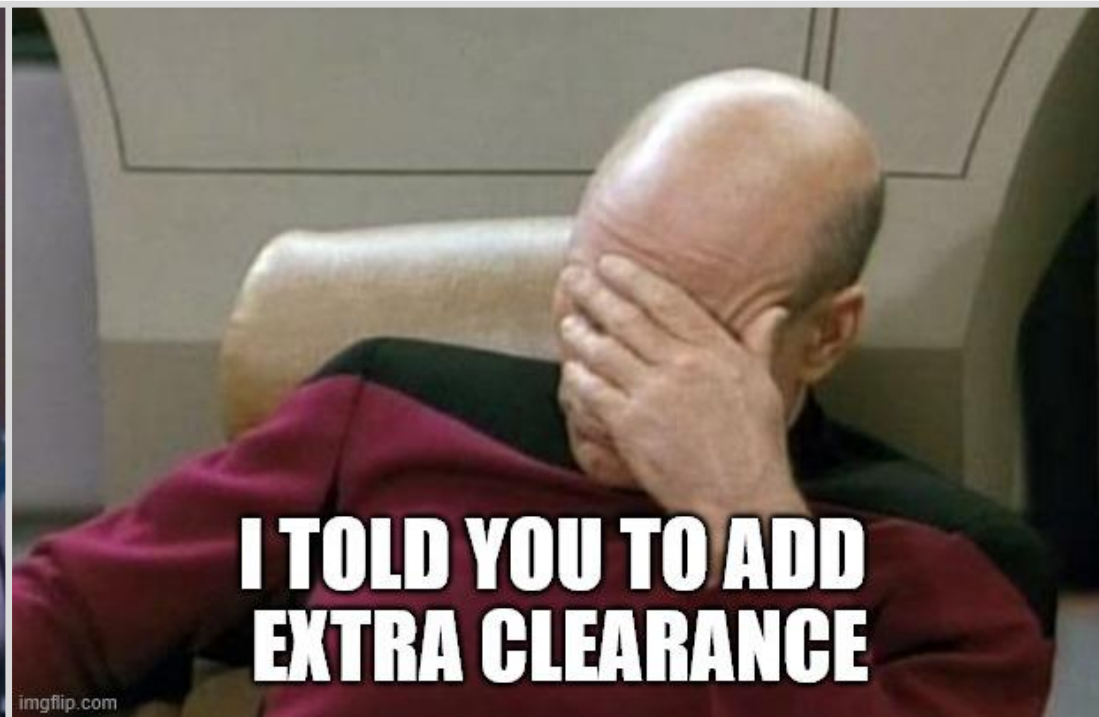
- How much XY clearance?
- It depends, but not very much for modern printers.
- Start with 0.25 mm.
- *Example:* Making a hole for a bolt.
 - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
 - Add 0.4 - 0.8 mm of clearance for a loose fit.

Adding Z Clearance

- How much Z clearance?
- It depends ...
- One layer or less for most situations.
- Increase clearance for:
 - Sagging extrusions.
 - Rough bottom surface from supports.
 - High and low spots on print bed.
 - Tilted print bed.

Summary

- If your project has assembled parts ...
 - You sometimes need scale factor.
 - You always need constant offset. Start with 0.25 mm for XY, one layer for Z.



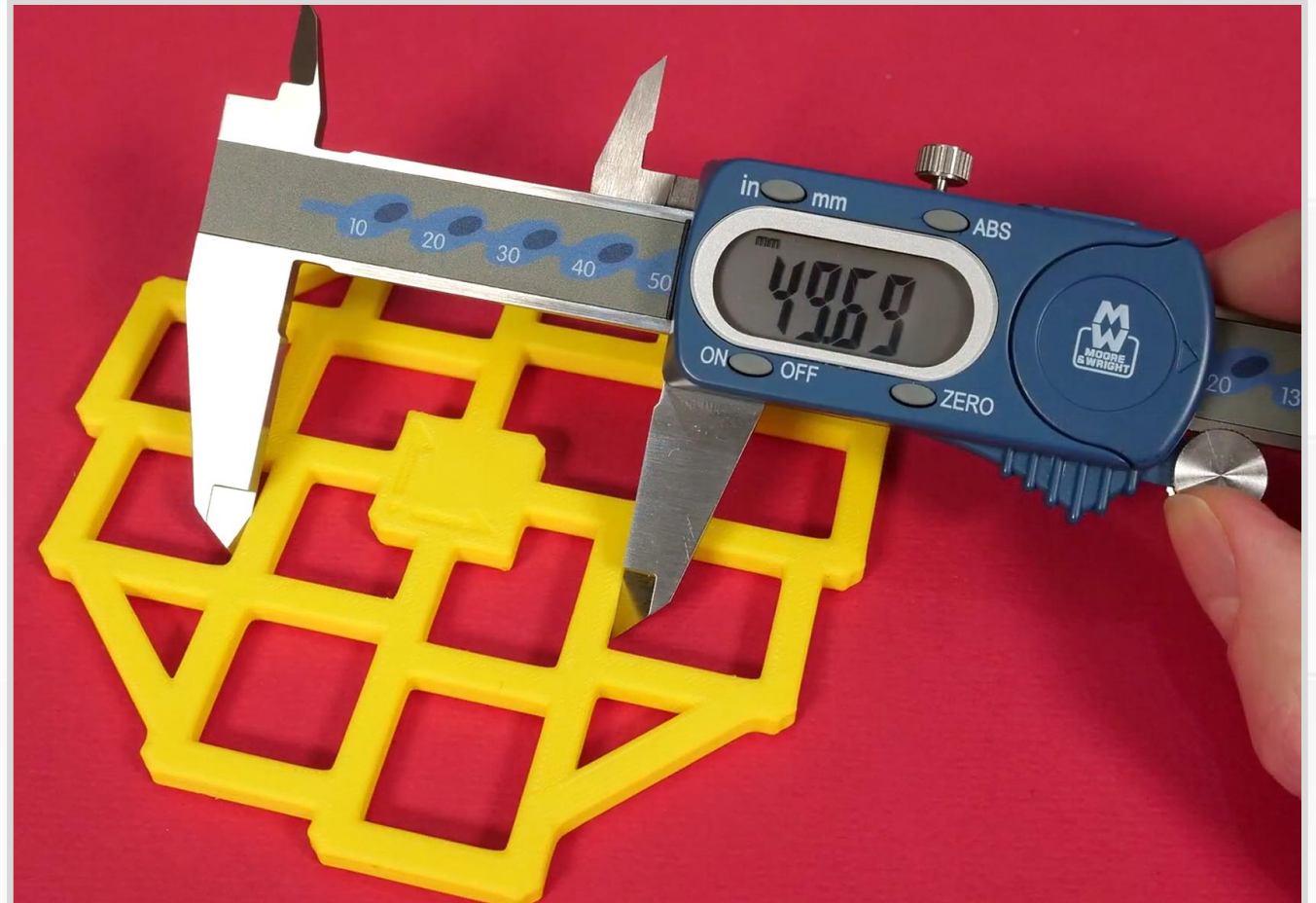
Calibration Prints

- Calibration cubes.
 - 20 mm cube.
 - Labels for X and Y-axis.
 - Tells you constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
 - Print-in-place clearance/tolerance.
 - Alters the gap between pieces.
 - Feel which pieces move freely.



Calibration Prints

- Califlower by Vector3D.
 - Walkthrough and calculator.
 - Inner dimensions.
 - Outer dimensions.
 - Shrinkage.
 - Skew. (If your rectangles turn into parallelograms).



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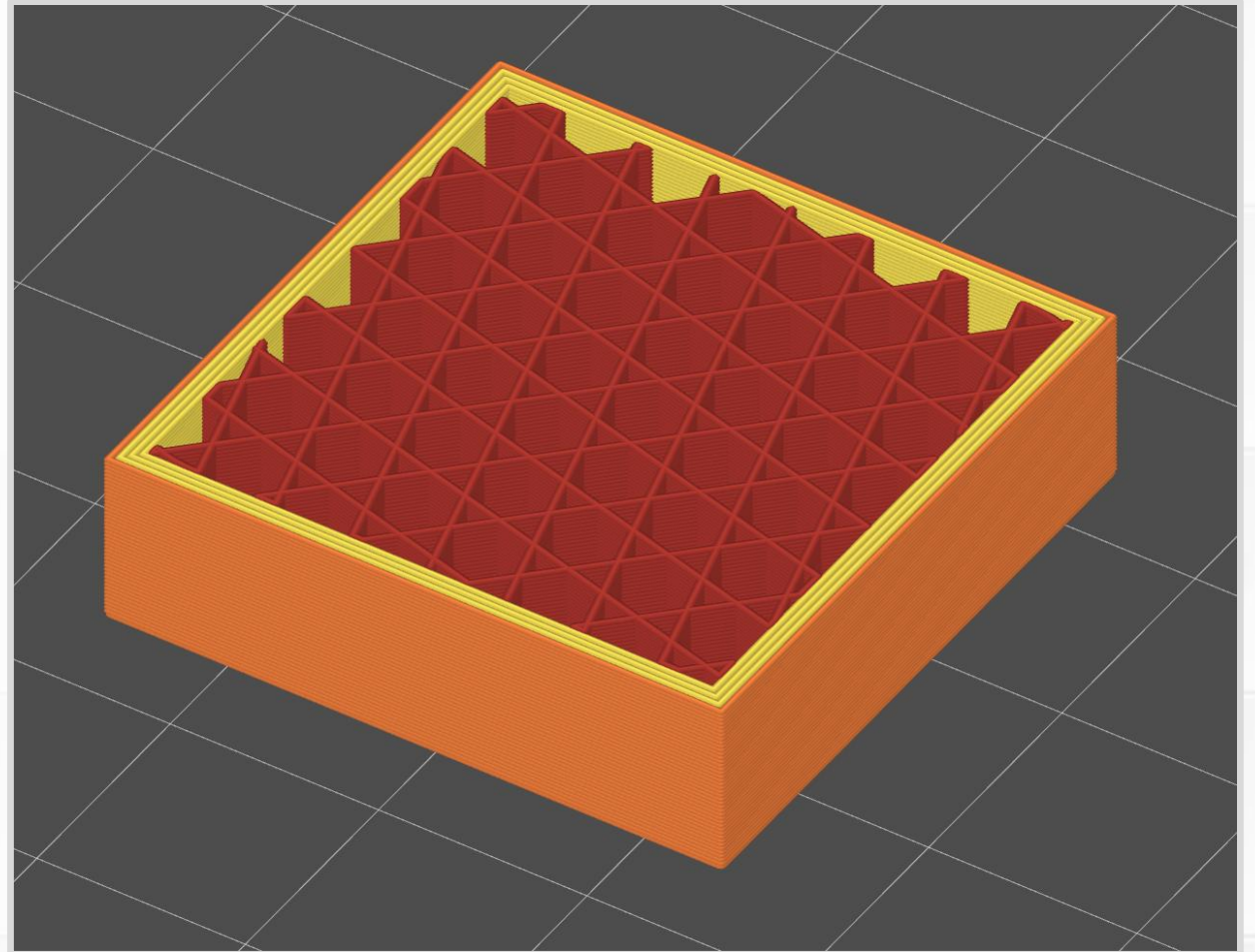
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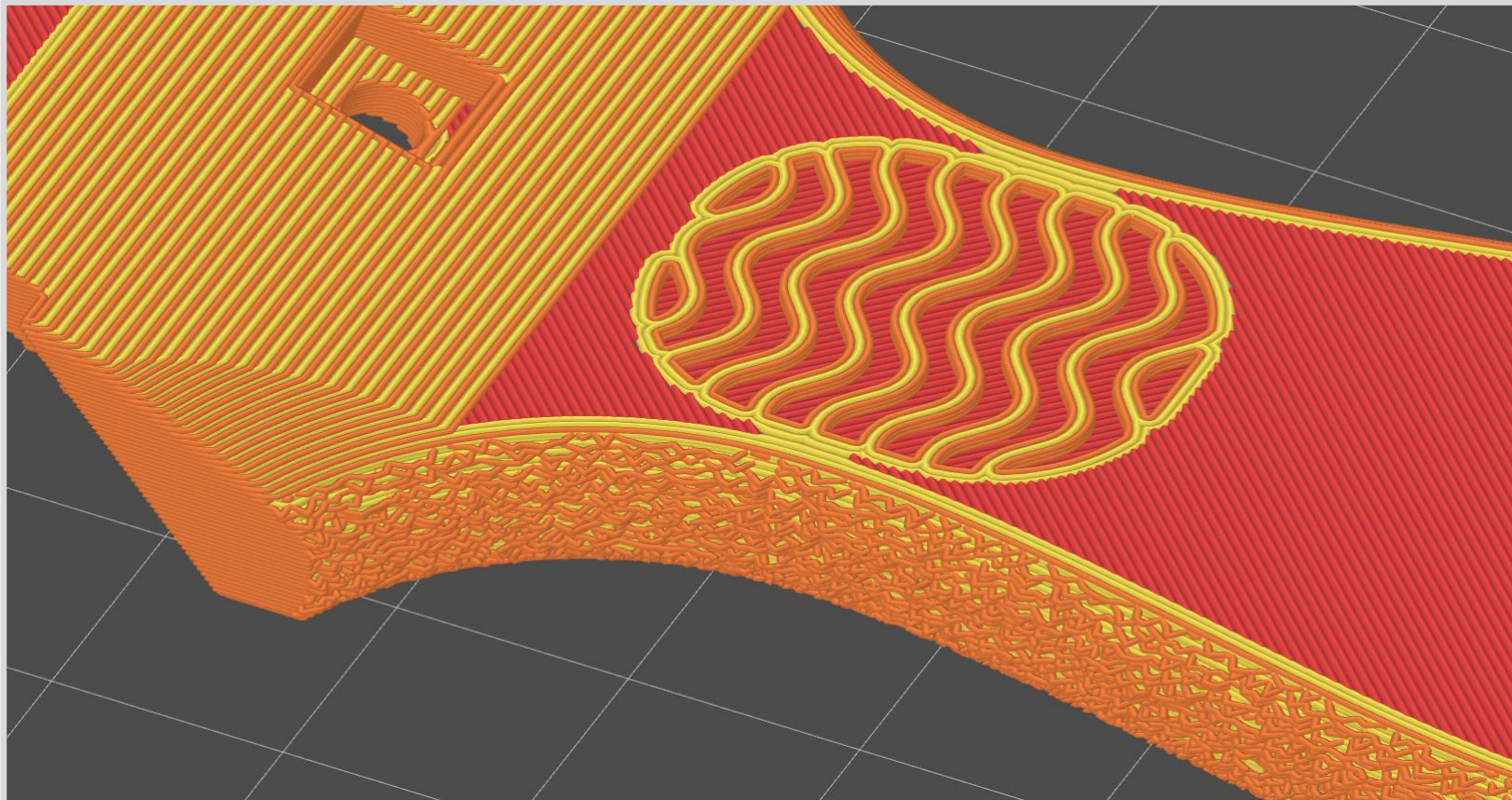
FFF Advantages

- Printers use tiny “building blocks.”
- Typical extrusions:
 - **0.2 mm tall** (1-2 pieces of paper).
 - **0.4 mm wide**.
- Not limited to uniform prints.
 - *Example:* Solid exterior and sparse infill.
- Great at making hollow objects.



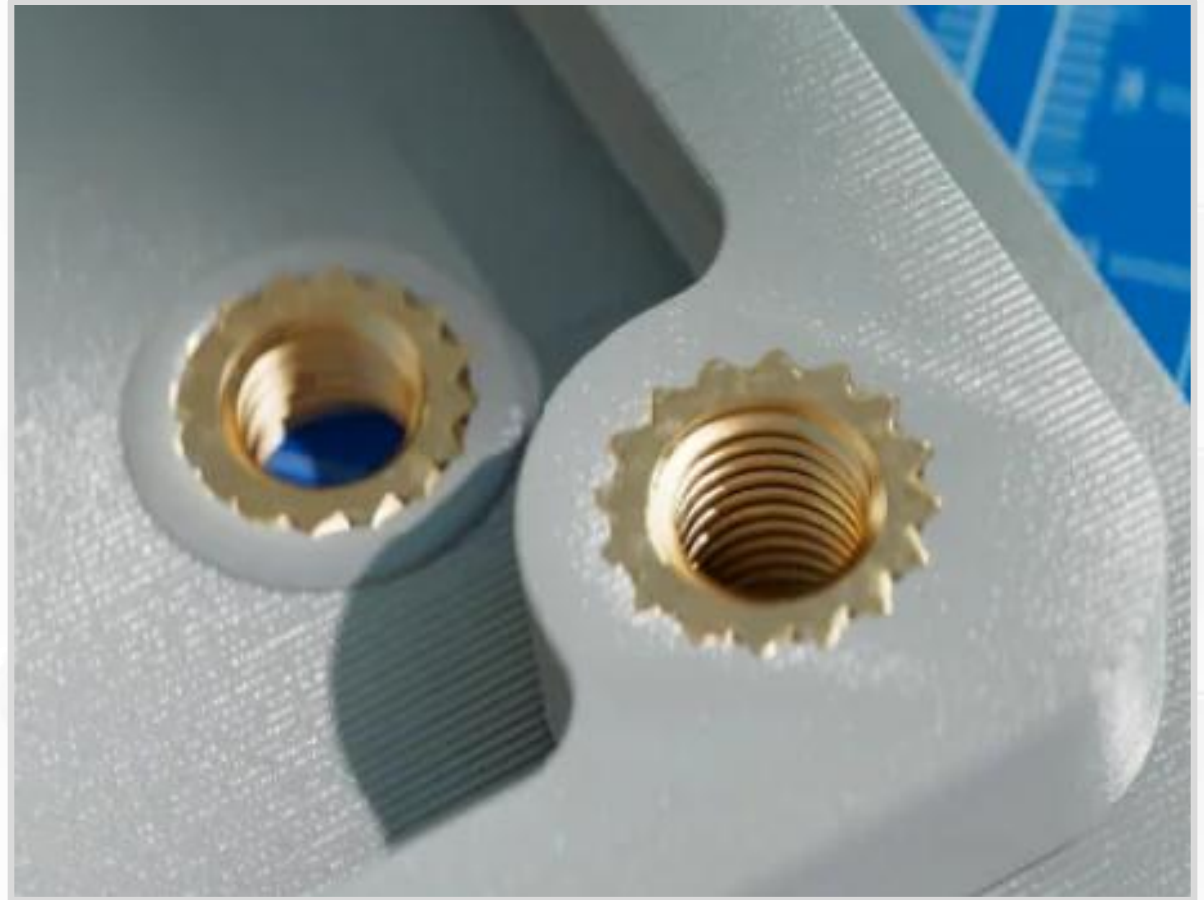
Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Plastic is soft → Screws easily wear out plastic holes.
- Threaded anchor for screws.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



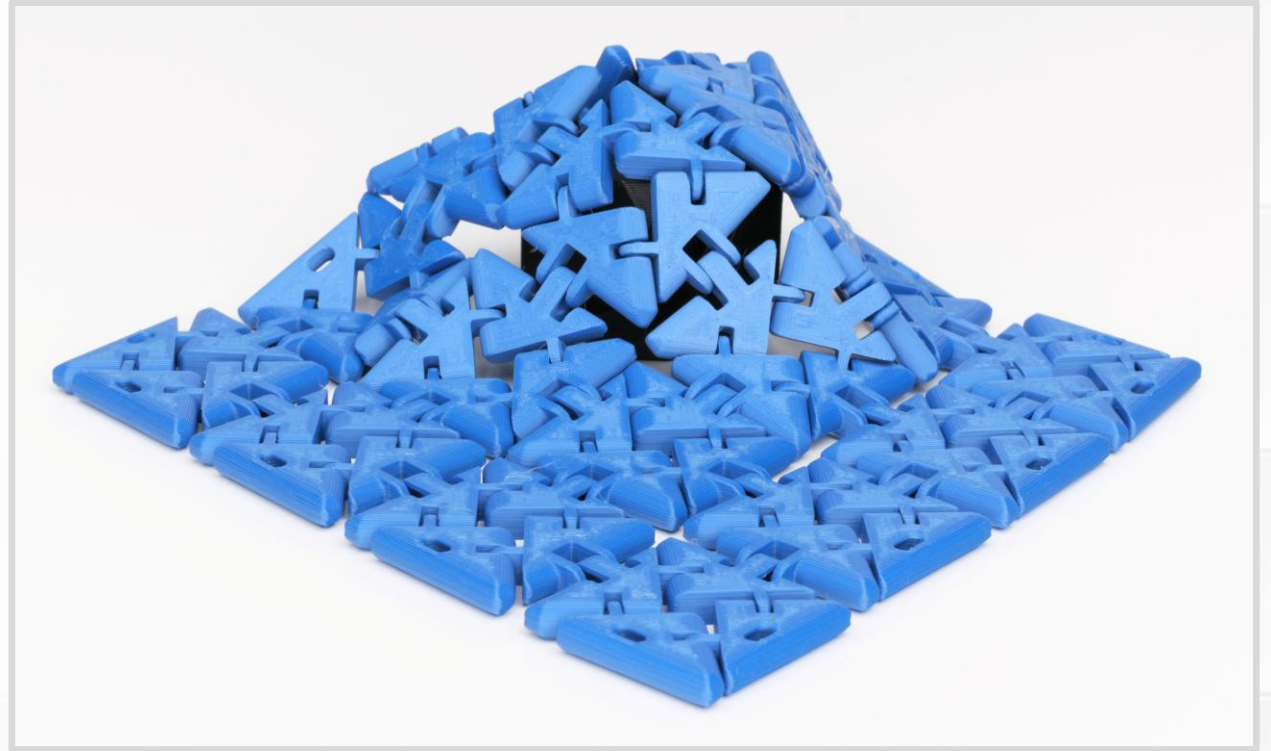
(Pictures from [CNC Kitchen](#))

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
 - Nuts and fasteners.
 - Magnets.
 - Weights.
 - Reinforcing plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
 - Need to be at the printer at the right layer.

Print-in-Place (PIP)

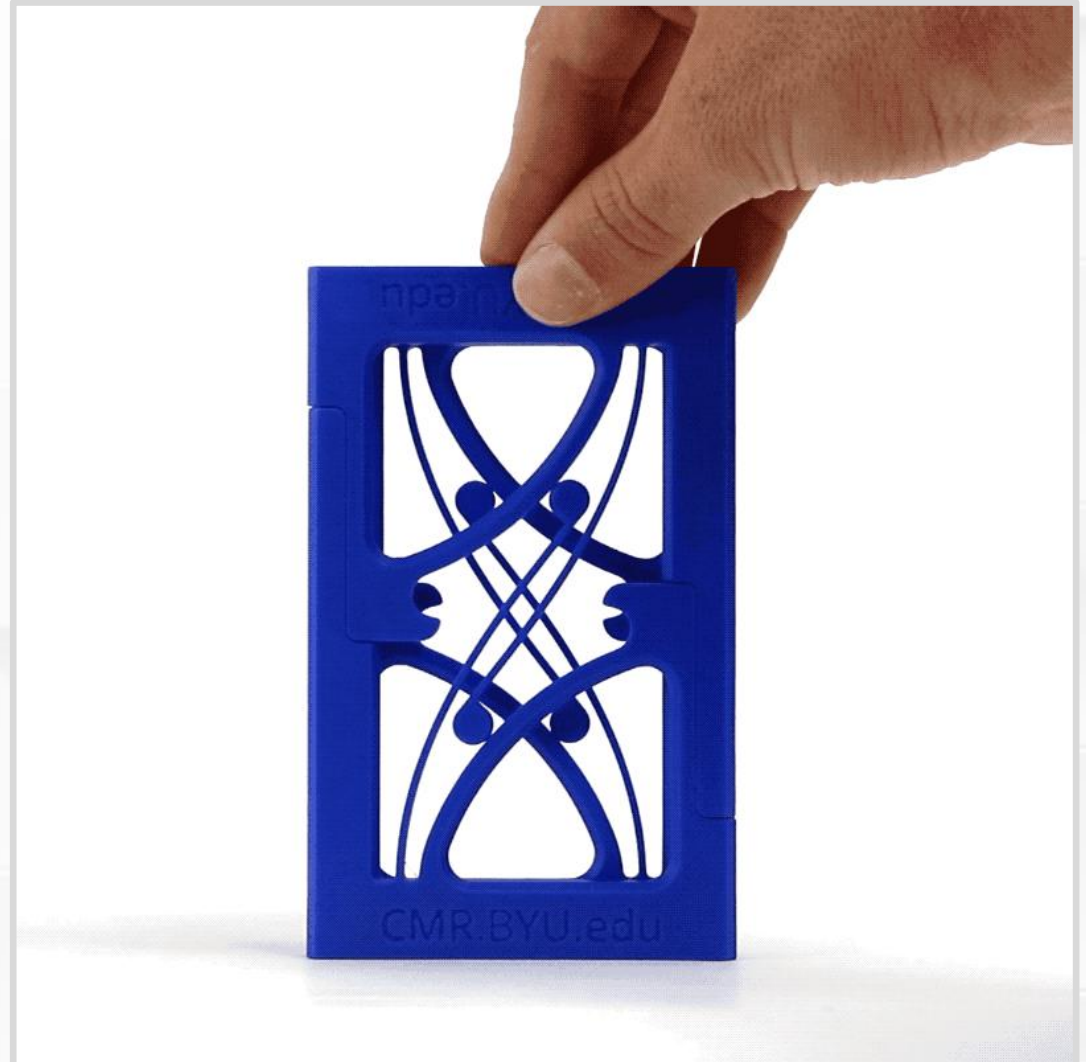
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by [printables.com/@Flying Gyroscope](https://printables.com/@FlyingGyroscope)



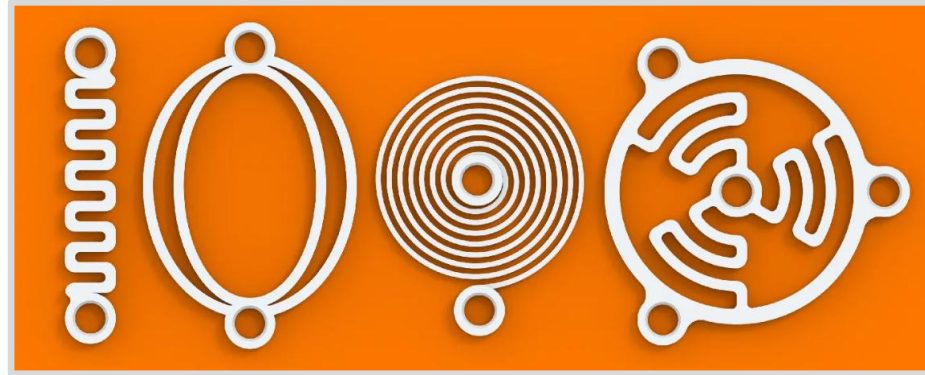
Functional Flexing

- Springs and **compliant mechanisms**.
- Features/shapes that function by bending and flexing.
- Elastic materials.
 - Springy.
 - No permanent deformation.
- Remember strength and layer lines.
 - Flex springs in XY plane for maximum durability.

(Animation, Constant Force
Mechanism by BYU CMR)



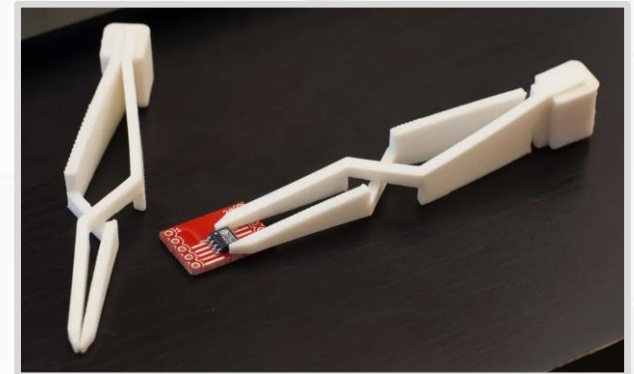
Springs by VC Design



Clip by AE



Cross tweezer by joo



Living hinge by DrJones

